

Sales Performance Dashboard

Project Report

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Tools: Excel | SQL | Power BI

1. Executive Summary

This report documents the Sales Performance Dashboard project, an end-to-end data analytics exercise built on the Sample Superstore dataset. The project moves through three connected stages — exploratory analysis in Excel, structured querying in SQL, and interactive visualization in Power BI — to convert raw transactional sales records into business-ready insights.

At a high level, the analysis shows total sales of 2.30M with a profit margin of 12.47%, led by the West region and the Technology category. It also surfaces a key risk area: several sub-categories (Tables, Bookcases, Supplies) generate sales but lose money, which is exactly the kind of pattern a business needs flagged early.

The sections that follow walk through the objectives, dataset, methodology, and findings in detail, followed by the challenges encountered, the project's current limitations, possible future enhancements, and the skills it demonstrates.

2. Project Overview

The Sales Performance Dashboard is a data analytics project developed to analyze sales, profit, customer, product, and regional performance using the Sample Superstore dataset.

The project demonstrates an end-to-end data analytics workflow, starting from data preparation and analysis in Microsoft Excel, followed by SQL-based analysis, and finally the development of an interactive Power BI dashboard.

The main goal of this project is to transform raw sales data into meaningful business insights that can help understand sales trends, identify profitable and underperforming areas, and support data-driven business decisions.

Project Workflow

Raw Dataset → Data Preparation → Excel Analysis → SQL Analysis → Power BI Dashboard → Business Insights

Technologies Used

- Microsoft Excel — Data preparation and exploratory analysis
- SQL — Business data analysis and querying
- Power BI — Interactive dashboard and visualization

- CSV — Source dataset format

3. Business Objectives

The primary objective of this project is to analyze the Superstore sales data and identify important patterns in sales and profitability.

- Analyze overall sales and profit performance
- Identify the best-performing and underperforming product categories and sub-categories
- Evaluate sales and profit performance across different regions and geographical areas
- Understand customer and order-level performance
- Identify products or areas generating high sales but relatively low profit
- Analyze sales performance over time to identify important trends and patterns
- Provide clear and interactive visualizations that make business performance easier to understand
- Generate actionable insights that can support data-driven business decisions

Key Business Questions

- How are overall sales and profit performing?
- Which categories and sub-categories contribute the most to sales and profit?
- Which regions perform best and which require improvement?
- Which products or areas have strong sales but weak profitability?
- How does sales performance change over time?
- Where are the major opportunities for improving profitability?

4. Dataset Description

The project uses the Sample Superstore dataset, which contains transactional sales data for a retail business. The dataset provides information about orders, customers, products, sales, discounts, profits, and geographical locations.

Dataset File

File: SampleSuperstore.csv

Main Data Fields

Field	Description
Row ID	Unique identifier for each row
Order ID	Unique identifier for an order
Order Date	Date on which the order was placed
Ship Date	Date on which the order was shipped
Ship Mode	Shipping method used for the order
Customer ID	Unique customer identifier
Customer Name	Name of the customer
Segment	Customer segment

Field	Description
Country	Country of the customer
City	City of the customer
State	State of the customer
Postal Code	Postal code of the customer
Region	Geographical region
Product ID	Unique product identifier
Category	Main product category
Sub-Category	Product sub-category
Product Name	Name of the product
Sales	Sales amount
Quantity	Number of units ordered
Discount	Discount applied to the order
Profit	Profit generated from the order

Dataset Usage

The same dataset was used throughout the project to maintain consistency between the different analytical stages: CSV Dataset → Excel → SQL → Power BI.

5. Data Cleaning & Preparation

Before performing the analysis, the dataset was reviewed and prepared to ensure that the data could be used consistently across Excel, SQL, and Power BI.

Data Preparation Steps

- Reviewed the dataset structure and available columns
- Checked the data for missing or inconsistent values
- Verified the data types of important fields such as dates, sales, quantity, discount, and profit
- Checked the dataset for duplicate records
- Reviewed categorical fields such as Category, Sub-Category, Segment, Region, and Ship Mode for consistency
- Ensured that numerical fields were suitable for calculations and aggregation
- Verified date fields for time-based analysis
- Prepared the cleaned dataset for further analysis in Excel, SQL, and Power BI

Data Flow

Prepared Dataset → Excel Analysis → SQL Analysis → Power BI Dashboard

This approach helped maintain consistency in the calculations and insights generated throughout the project.

6. Excel Analysis

Microsoft Excel was used as an initial analytical tool to explore the Superstore dataset and understand the overall sales and profitability performance.

Excel Analysis Activities

- Reviewed and organized the Superstore dataset
- Analyzed sales and profit performance
- Examined product categories and sub-categories
- Compared performance across different regions
- Analyzed customer segments and their contribution to sales
- Reviewed order quantities, discounts, sales, and profit
- Used Excel-based calculations and analysis to identify important business patterns
- Prepared the data and analytical understanding required for the Power BI dashboard

Purpose of Excel Analysis

- Understand the structure and characteristics of the dataset
- Perform initial exploratory analysis
- Identify important sales and profitability patterns
- Validate the data before performing SQL analysis and dashboard development

The Excel workbook is available in: [Excel/SampleSuperstore.xlsx](#)

7. SQL Analysis

SQL was used to perform structured analysis of the Superstore dataset and answer key business questions related to sales, profit, customers, products, and geographical performance.

SQL Analysis Areas

- Calculating overall sales and profit performance
- Analyzing sales and profit by category and sub-category
- Comparing performance across different regions
- Analyzing customer segments
- Identifying top-performing products
- Identifying products with lower or negative profitability
- Analyzing the relationship between sales, discounts, quantity, and profit
- Examining sales performance over time
- Ranking products and business areas based on sales and profitability

SQL Techniques Used

- SELECT, WHERE, GROUP BY, ORDER BY
- Aggregate functions such as SUM(), COUNT(), and AVG()
- Filtering and sorting data
- Group-level business analysis

- Ranking and comparison of results

The SQL queries are available in: SQL/Sales_Analysis.sql

Sample Query

Example of a query used to identify profit performance by sub-category:

```
SELECT "Sub-Category", SUM(Sales) AS Total_Sales, SUM(Profit) AS Total_Profit
FROM superstore
GROUP BY "Sub-Category"
ORDER BY Total_Profit DESC;
```

8. Power BI Dashboard

Power BI was used to transform the analyzed Superstore data into an interactive dashboard for monitoring sales and profitability performance.

Dashboard Objectives

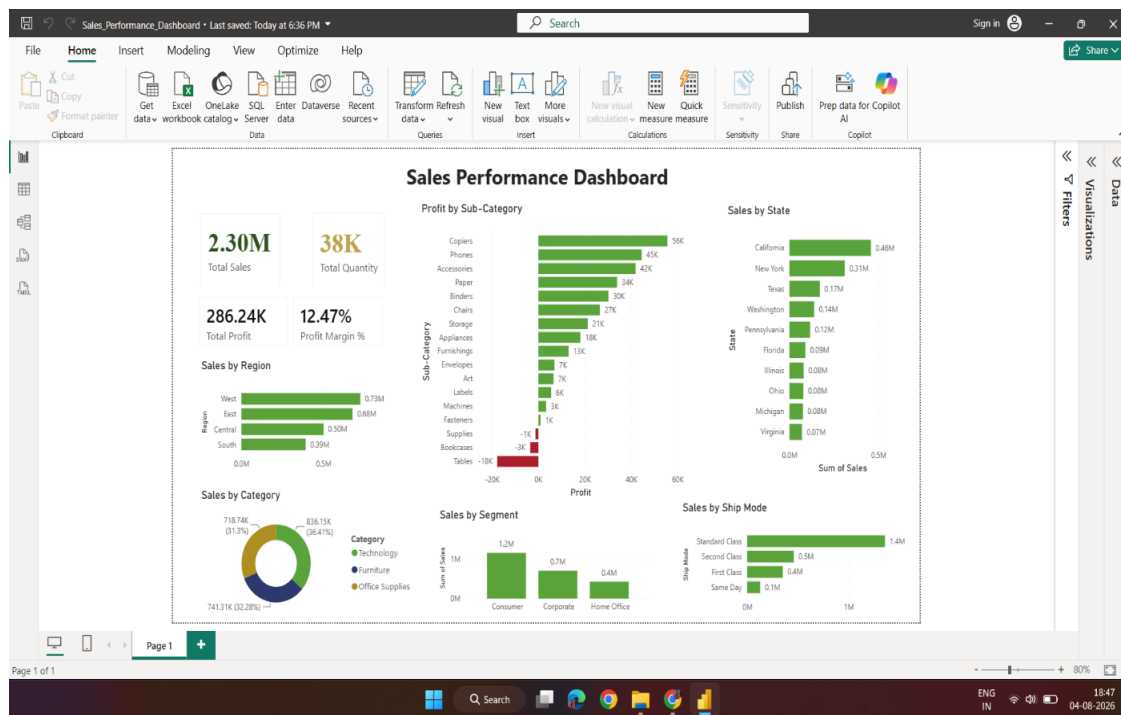
- Provide a quick overview of overall business performance
- Monitor important sales and profitability metrics
- Compare performance across product categories and sub-categories
- Analyze regional performance
- Understand customer segment performance
- Identify trends and patterns in sales and profit
- Make business insights easier to interpret through interactive visualizations

Dashboard Components

- Sales Performance
- Profit Performance
- Product Performance
- Category & Sub-Category Analysis
- Regional Performance
- Customer Segment Analysis
- Time-Based Performance

The Power BI report is available in: PowerBI/Sales_Performance_Dashboard.pbix

Dashboard Preview



9. Key Performance Indicators (KPIs)

The Power BI dashboard presents four key performance indicators that provide a quick overview of the overall business performance.

KPI	Value	Description
Total Sales	2.30M	Total revenue generated from sales
Total Quantity	38K	Total number of products sold
Total Profit	286.24K	Total profit generated from sales
Profit Margin	12.47%	Percentage of sales retained as profit

KPI Interpretation

- Total Sales of 2.30M indicates the overall sales generated across all orders in the dataset
- Total Quantity of 38K represents the total number of units sold
- Total Profit of 286.24K represents the overall profit generated after accounting for the associated costs
- Profit Margin of 12.47% indicates the proportion of sales that was converted into profit

10. Dashboard Visual Analysis

The Power BI dashboard contains multiple visualizations that provide detailed insights into sales and profitability across regions, products, states, customer segments, categories, and shipping methods.

10.1 Sales by Region

The Sales by Region chart compares total sales across the four major regions.

Region	Sales
West	0.73M
East	0.68M
Central	0.50M
South	0.39M

The West region records the highest sales, while the South region has the lowest sales among the four regions.

10.2 Profit by Sub-Category

The Profit by Sub-Category visualization compares profitability across different product sub-categories.

Highest-profit sub-categories:

- Copiers — 56K
- Phones — 45K
- Accessories — 42K
- Paper — 34K
- Binders — 30K

Sub-categories generating negative profit:

- Tables — approximately -18K
- Bookcases — approximately -3K
- Supplies — approximately -1K

This visualization highlights that high sales do not necessarily result in high profitability and helps identify sub-categories that may require further investigation.

10.3 Sales by State

The Sales by State chart highlights the states contributing the highest sales.

- California — 0.46M
- New York — 0.31M
- Texas — 0.17M
- Washington — 0.14M
- Pennsylvania — 0.12M

California is the highest-performing state by sales in the dashboard.

10.4 Sales by Category

The Sales by Category donut chart shows the contribution of the three major product categories.

Category	Sales	Share
Technology	836.15K	36.41%
Furniture	741.31K	32.28%
Office Supplies	718.74K	31.31%

Technology contributes the largest share of total sales, followed by Furniture and Office Supplies. The distribution is relatively balanced across the three categories.

10.5 Sales by Segment

The Sales by Segment chart compares sales generated by different customer segments.

Segment	Sales
Consumer	1.2M
Corporate	0.7M
Home Office	0.4M

The Consumer segment generates the highest sales, making it the largest customer segment in terms of revenue contribution.

10.6 Sales by Ship Mode

The Sales by Ship Mode chart compares sales across different shipping methods.

Ship Mode	Approx. Sales
Standard Class	1.4M
Second Class	0.5M
First Class	0.4M
Same Day	0.1M

Standard Class accounts for the largest share of sales, while Same Day shipping contributes the smallest sales volume.

Overall Dashboard Analysis

The visualizations provide a consolidated view of business performance and make it easier to identify:

- High-performing regions and states
- Strong and weak product sub-categories
- Category-level sales contribution
- The most important customer segment
- Customer preference for different shipping methods
- Areas where strong sales may not translate into strong profitability

11. Challenges Faced & Solutions

Working across three tools on the same dataset surfaced a few practical challenges, along with how each was addressed:

Maintaining consistency across Excel, SQL, and Power BI

Since the same dataset was used at every stage, ensuring that cleaned values, column names, and calculations matched across all three tools required careful cross-checking after each step.

Handling negative-profit records

Some sub-categories showed sales with negative profit due to high discounting. Identifying and correctly attributing these required filtering and grouping the data by sub-category rather than looking at overall totals.

Structuring SQL queries for business questions

Translating open-ended business questions (e.g., "which areas have high sales but low profit") into precise GROUP BY and aggregate queries took some iteration to get the right level of granularity.

Designing a dashboard that stays readable

With multiple dimensions (region, category, segment, ship mode, time), keeping the Power BI dashboard visually clear without overcrowding it required choosing a focused set of visuals rather than including every possible chart.

12. Limitations

- The analysis is based on a static sample dataset rather than live or continuously updated sales data
- The dataset covers a fixed historical period, so findings reflect past performance and may not capture recent market shifts
- Profitability drivers (e.g., discount strategy, shipping cost) are observed from the data but not modeled with external cost or pricing data
- The dashboard is designed for exploration and reporting, not for predictive forecasting

13. Future Enhancements

- Automate the pipeline so new sales data refreshes the Power BI dashboard without manual reprocessing
- Add time-series forecasting to project future sales and profit trends
- Incorporate discount-sensitivity analysis to recommend optimal discount thresholds by sub-category
- Extend the SQL layer into a proper data warehouse or use a cloud database instead of flat CSV files
- Add customer-level segmentation (e.g., RFM analysis) to complement the current product- and region-level views
- Publish the dashboard to Power BI Service for shareable, web-based access

14. Skills Demonstrated

- Data cleaning and preparation for multi-tool analytical workflows
- Exploratory data analysis using Microsoft Excel
- Business-focused SQL querying: aggregation, filtering, grouping, and ranking
- Dashboard design and interactive data visualization in Power BI
- Translating raw data into business insights and recommendations

- End-to-end project documentation and reporting

15. Conclusion

The Sales Performance Dashboard project successfully demonstrates a complete data analytics workflow — from raw data to business insight — using Excel, SQL, and Power BI. The analysis identified the highest-performing regions, states, categories, and segments, while also surfacing an important risk: several sub-categories generate sales without generating profit.

Beyond the specific findings, the project shows the ability to work with a dataset consistently across multiple tools, translate business questions into structured analysis, and communicate results through a clear, interactive dashboard — the core skill set expected of a data analyst.

16. References & Data Source

- Dataset: Sample Superstore dataset (SampleSuperstore.csv)
- Tools used: Microsoft Excel, SQL, Power BI